Math 2551 Worksheet: Double Integrals on Rectangles

1. Without using an iterated integral, evaluate the double integral $\iint_R (4-2y)\ dA$, $R = [0,1]\times[0,1]$, by identifying it as the volume of a solid.

2. The integral $\iint_R \sqrt{9-y^2}$, where $R = [0,4]\times[0,2]$, represents the volume of a solid. Sketch the solid.

3. Find $\displaystyle\iint_R \frac{xy^2}{x^2+1}\ dA, \qquad R\colon\ 0 \le x \le 1,\ -3 \le y \le 3$

4. Find the volume of the region bounded above by the elliptical paraboloid $z = 16 - x^2 - y^2$ and below by the square $R: 0 \le x \le 2, 0 \le y \le 2$.

5. Use Fubini's Theorem to evaluate the integral
$$\int_0^1 \int_0^3 xe^{xy}\ dx\ dy.$$

6. **Challenge**: Use a trig substitution $y = x\tan(\theta)$ or $x = y\tan(\theta)$ to show that the iterated integrals
$$\int_0^1 \int_0^1 \frac{x^2-y^2}{(x^2+y^2)^2}\ dy\ dx \qquad \text{and} \qquad \int_0^1 \int_0^1 \frac{x^2-y^2}{(x^2+y^2)^2}\ dx\ dy$$
are not equal. Why does this not violate Fubini's Theorem?